

Daily Geometry Drill #7 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. Write down the locus of points whose distance from the origin is 5.

Answer: $x^2 + y^2 = 25$

Method: $\sqrt{x^2 + y^2} = 5$ rearranges to $x^2 + y^2 = 25$, a circle of radius 5.

Investigate further: Describe the set of points whose distance from the origin is at least 3 and at most 5, and find its area.

2. Write down the locus of points equidistant from the two points $(0, 0)$ and $(4, 0)$.

Answer: $x = 2$

Method: Squaring $x^2 + y^2 = (x - 4)^2 + y^2$: $x^2 = (x - 4)^2$, so $x = 2$.

Investigate further: Now take the points $(0, 0)$ and $(4, 2)$. Find the equation of the locus, and check that it passes through their midpoint.

3. The locus of P with $PA = 2PB$, where $A = (0, 0)$ and $B = (6, 0)$, is a circle. It crosses the segment AB at one point. Write down the x -coordinate of that point.

Answer: 4

Method: On the x -axis between A and B , $PA = x$ and $PB = 6 - x$, so $x = 2(6 - x)$, giving $x = 4$. The circle's other crossing is $x = 12$, from $x = 2(x - 6)$.

Investigate further: The circle also crosses the x -axis at $x = 12$. Use the two crossings to write down its centre and radius without any algebra.

4. Write down the locus of points that are always 3 units from the fixed line $y = 4$.

Answer: $y = 1$ or $y = 7$

Method: $|y - 4| = 3$ gives $y = 1$ and $y = 7$, two parallel lines 3 units from $y = 4$.

Investigate further: What is the locus of points 3 units from the circle $x^2 + y^2 = 16$?

5. The locus of points whose distance to the point $(0, 1)$ equals their distance to the line $y = -1$ is a parabola. Write down its equation.

Answer: $x^2 = 4y$

Method: $\sqrt{x^2 + (y - 1)^2} = |y + 1|$. Squaring, $x^2 + y^2 - 2y + 1 = y^2 + 2y + 1$, so $x^2 = 4y$.

Investigate further: Move the line to $y = -3$, keeping the point $(0, 1)$. Find the new equation and the vertex of the parabola.

6. The locus of points whose distance to the point $(1, 2)$ is exactly 3 is a circle. Write down its equation.

Answer: $(x - 1)^2 + (y - 2)^2 = 9$

Method: Points at distance exactly 3 from $(1, 2)$ form the circle centred there with radius 3.

Investigate further: Expand the equation into the form $x^2 + y^2 + 2gx + 2fy + c = 0$. The constant is $c = 1 + 4 - 9 = -4$; what does $c < 0$ tell you about where the origin is?

7. Write down the locus of the midpoint M of every radius OQ of the circle $x^2 + y^2 = 4$ as Q runs around the circle (with O the origin).

Answer: $x^2 + y^2 = 1$

Method: Q runs over $x^2 + y^2 = 4$ (radius 2); the midpoint $M = OQ/2$ scales radii by $\frac{1}{2}$, giving $x^2 + y^2 = 1$.

Investigate further: Now let M be the midpoint of PQ , where $P = (4, 0)$ is fixed. Show that the locus is still a circle, and find its centre and radius.

8. The equation $x^2 + y^2 = 2x$ describes a circle. Write down its radius.

Answer: 1

Method: $x^2 + y^2 = 2x$ rearranges to $(x - 1)^2 + y^2 = 1$, radius 1.

Investigate further: Show that every circle $x^2 + y^2 = 2ax$ ($a \neq 0$) passes through the origin and touches the y -axis there.

9. Write down whether the statement “ $x^2 + y^2 = 4$ implies $x = \pm 2$ or $y = \pm 2$ ” is true or false.

Answer: false

Method: The point $(\sqrt{2}, \sqrt{2})$ lies on $x^2 + y^2 = 4$ yet has $x \neq \pm 2$ and $y \neq \pm 2$; the implication fails in both coordinates.

Investigate further: Is it true that $x^2 + y^2 = 4$ implies $|x| \leq 2$ and $|y| \leq 2$? Is the converse true?

10. Write down the locus of points P with $PA = PB$, where $A = (1, 3)$ and $B = (5, 3)$. (Give its equation in x .)

Answer: $x = 3$

Method: A and B are at equal height $y = 3$, so the perpendicular bisector is the vertical line $x = \frac{1+5}{2} = 3$.

Investigate further: Find the locus of P with $PA = PB$ for $A = (1, 3)$ and $B = (5, 7)$, and check that it passes through the midpoint of AB .

Section B Manipulation Drills

1. The locus of P with $PA = 3PB$, where $A = (0, 0)$ and $B = (8, 0)$, is a circle. What is its radius? (after *tmua.fyi Challenge Mock 1 Paper 1 Q14*)

- A) 9
- B) 6
- C) 3
- D) 8

Answer: C) 3

Method: On the x -axis, $|x| = 3|x - 8|$ gives $x = 6$ and $x = 12$, the ends of a diameter, so the centre is $(9, 0)$ and the radius is 3. Check by squaring: $x^2 + y^2 = 9((x - 8)^2 + y^2)$ becomes $(x - 9)^2 + y^2 = 9$. Option A is the centre's x -coordinate; option B is the diameter; option D is AB .

Investigate further: What happens to the circle as the ratio 3 is replaced by numbers closer and closer to 1? What is the locus when the ratio is exactly 1?

2. Locus of points equidistant from $(0, 0)$ and $(0, 4)$ is?

- A) $x = 2$
- B) $y = 2$
- C) $y = 4$
- D) $x = 4$

Answer: B) $y = 2$

Method: $x^2 + y^2 = x^2 + (y - 4)^2$ gives $y = 2$, the horizontal bisector.

Investigate further: Show that for $A = (0, 0)$ and $B = (a, b)$ the locus of points equidistant from A and B is $ax + by = \frac{1}{2}(a^2 + b^2)$.

3. The points satisfying $x^2 + y^2 \leq 4$ form a filled disk. Its area is?

- A) 2π
- B) 4π
- C) 8π
- D) 16

Answer: B) 4π

Method: The disk has radius 2, so its area is $\pi r^2 = 4\pi$.

Investigate further: Find the area of the part of the disk $x^2 + y^2 \leq 4$ with $x + y \geq 2$.

4. The points $A = (0, 0)$ and $B = (4, 0)$ are fixed. The point P moves in the region $y > 0$ so that $\angle APB = 45^\circ$. What is the locus of P ?
- A) an arc of the circle with centre $(2, 0)$ and radius 2
 - B) an arc of the circle with centre $(2, -2)$ and radius $2\sqrt{2}$
 - C) part of the line $x = 2$
 - D) an arc of the circle with centre $(2, 2)$ and radius $2\sqrt{2}$

Answer: D) an arc of the circle with centre $(2, 2)$ and radius $2\sqrt{2}$

Method: A fixed chord AB subtends a fixed angle only along an arc (angles in the same segment are equal). The angle at the centre is $2 \times 45^\circ = 90^\circ$, so the radius is $\frac{AB}{\sqrt{2}} = 2\sqrt{2}$, and the centre is 2 from the midpoint $(2, 0)$ on the same side as P (the angle is acute), i.e. at $(2, 2)$. Option A is the 90° circle (Thales); on the upper arc of option B's circle, AB subtends $180^\circ - 45^\circ = 135^\circ$.

Investigate further: What is the locus if P may lie on either side of AB (still with $\angle APB = 45^\circ$)?

5. A point is known to lie strictly inside the unit disk $x^2 + y^2 < 1$. Which of the following statements is *necessarily* true for such a point?

- A) $x^2 + y^2 < \frac{1}{2}$
- B) $x + y < 1$
- C) $x^2 + y^2 < 2$
- D) $x^2 + y^2 > 0$

Answer: C) $x^2 + y^2 < 2$

Method: If $x^2 + y^2 < 1$ then certainly $x^2 + y^2 < 2$. Option A fails at $(0.8, 0)$; option B fails at $(0.7, 0.7)$, which is inside the disk but has $x + y = 1.4$; option D fails at the origin.

Investigate further: Option A is the other way round: show that $x^2 + y^2 < \frac{1}{2}$ is sufficient for the point to be inside the unit disk, but not necessary.

6. The Apollonius locus $PA = 3PB$ with $A = (0, 0)$, $B = (2, 0)$: what is the x -coordinate of its right endpoint on the x -axis?

- A) 3
- B) 4
- C) 6
- D) 9

Answer: A) 3

Method: On the x -axis $|x| = 3|x - 2|$ gives internal $x = \frac{3}{2}$ and external $x = 3$; the right endpoint is 3.

Investigate further: Use the two crossings $x = \frac{3}{2}$ and $x = 3$ to find the centre and radius of this circle.

7. To prove a point P lies on the unit circle $x^2 + y^2 = 1$, which single condition is *sufficient* on its own?

- A) $x^2 + y^2 < 1$
- B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$
- C) P is not the origin
- D) $x + y = 1$

Answer: B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$

Method: This point satisfies $x^2 + y^2 = \frac{1}{4} + \frac{3}{4} = 1$ exactly, so it pins P on the circle. Option A puts P strictly *inside*; C and D do not force $x^2 + y^2 = 1$.

Investigate further: Option D, $x + y = 1$, meets the circle at $(1, 0)$ and $(0, 1)$. Explain why it is neither necessary nor sufficient for P to lie on the circle.

8. Which property is necessary, but not sufficient, for a quadrilateral to be a rectangle?

- A) all four angles are right angles
- B) the diagonals are equal in length
- C) the diagonals are perpendicular
- D) the diagonals are equal in length and bisect each other

Answer: B) the diagonals are equal in length

Method: Every rectangle has equal diagonals, so B is necessary; an isosceles trapezium also has equal diagonals, so B is not sufficient. Options A and D are both necessary and sufficient. Option C is not even necessary: a 1×2 rectangle's diagonals are not perpendicular.

Investigate further: Find a property that is sufficient but not necessary for a quadrilateral to be a rectangle.

9. A circle has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Write down its centre.

Answer: $(2, 3)$

Method: In $x^2 + y^2 + 2gx + 2fy + c = 0$, the centre is $(-g, -f)$; with $2g = -4$ and $2f = -6$ we get $(2, 3)$.

Investigate further: The radius is 2. Show that this circle touches the y -axis. Does it touch the x -axis?

10. Write down the radius of the circle $x^2 + y^2 - 4x + 6y + 9 = 0$.

Answer: 2

Method: Completing the square: $(x - 2)^2 + (y + 3)^2 = (-9) + 4 + 9 = 4$, radius 2.

Investigate further: Only the sign of the y term differs from the circle in B9. How are the two circles placed, and how many common tangents do they have?

Section C Substitution & Structure

1. The locus of P with $PA = 2PB$, where $A = (-2, 0)$ and $B = (4, 0)$, is a circle. What is its centre?

- A) $(6, 0)$
- B) $(8, 0)$
- C) $(10, 0)$
- D) $(2, 0)$

Answer: A) $(6, 0)$

Method: Square $PA = 2PB$: $(x+2)^2 + y^2 = 4((x-4)^2 + y^2)$, so $x^2 + 4x + 4 + y^2 = 4x^2 - 32x + 64 + 4y^2$, i.e. $x^2 - 12x + 20 + y^2 = 0$, or $(x-6)^2 + y^2 = 16$. Centre $(6, 0)$, radius 4. Faster: the x -axis crossings from $|x+2| = 2|x-4|$ are $x = 2$ and $x = 10$, and the centre is their midpoint. Option C is the far crossing; option D is the near one.

Investigate further: For which ratio k does the locus of $PA = kPB$, with the same A and B , pass through the origin?

2. The locus of points equidistant from the lines $L_1 : x + y = 0$ and $L_2 : x - y = 0$ is:

- A) $y = x$
- B) $y = -x$
- C) $y = |x|$
- D) $x = 0$ or $y = 0$

Answer: D) $x = 0$ or $y = 0$

Method: The distances to L_1 and L_2 are $\frac{|x+y|}{\sqrt{2}}$ and $\frac{|x-y|}{\sqrt{2}}$. Setting them equal and squaring: $(x+y)^2 = (x-y)^2$, so $4xy = 0$, i.e. $x = 0$ or $y = 0$: the two bisectors of the angles between the lines. Options A and B are the given lines themselves; option C is half of each.

Investigate further: Find the locus of points equidistant from the line $y = 2x$ and the x -axis, and check that it consists of two perpendicular lines.

3. *Claim.* If A , B and P lie on a circle with centre O , and P is on the major arc AB , then $\angle AOB = 2\angle APB$.

Proof. Draw the diameter PD . Triangles OPA and OPB are isosceles, so by the exterior angle theorem $\angle AOD = 2\angle APO$ and $\angle BOD = 2\angle BPO$. Adding, $\angle AOB = 2(\angle APO + \angle BPO) = 2\angle APB$.

Which statement about this argument is correct?

- A) It is correct and covers every position of P on the major arc.
- B) The exterior angle step fails, because $\angle OAP \neq \angle OPA$ in general.
- C) Adding assumes O lies inside $\angle APB$; when it does not, the angles must be subtracted, so that case is unproved.
- D) The claim itself is false when O lies outside triangle APB .

Answer: C)

Method: The claim is true (it is the angle at the centre theorem), but the proof only covers one case. If D lies on the minor arc AB , then O is inside $\angle APB$ and the two parts add. If P is near A and the minor arc is short, D lies on the major arc and O is outside $\angle APB$. Then $\angle APB$ is the difference of $\angle APO$ and $\angle BPO$, and $\angle AOB$ is the difference of $\angle AOD$ and $\angle BOD$, so the same two exterior-angle facts give $\angle AOB = 2\angle APB$ by subtraction, a step the proof never takes. Option B is wrong because $OA = OP$ makes $\angle OAP = \angle OPA$; option D is wrong because the claim holds in both cases.

Investigate further: Write out the subtraction case in full. Then say what happens when O lies on PA itself.

4. $PQRS$ is a quadrilateral. Consider the conditions: I. its diagonals bisect each other; II. its diagonals are equal in length; III. its diagonals are perpendicular. Which is the smallest set of these conditions that together guarantee $PQRS$ is a square? (after TMUA 2020 Paper 2 Q7)
- A) I and II
 B) I and III
 C) II and III
 D) I, II and III
 E) no set of them is enough

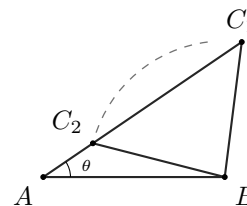
Answer: D) I, II and III

Method: I alone makes a parallelogram. I and II give a rectangle; I and III give a rhombus; neither need be a square. II and III without I fail too: diagonals of length 2 crossing at right angles but not at their midpoints give a kite that is not a square. All three together give a parallelogram that is both a rectangle and a rhombus, which is a square.

Investigate further: Which single one of the three conditions, added to “ $PQRS$ is a rectangle”, guarantees a square?

5. In $\triangle ABC$, $AB = 8$, $BC = 6$ and $\angle A = \theta$. Exactly two non-congruent triangles fit these data, and the larger has four times the area of the smaller. What is $\cos \theta$? (after TMUA 2018 Paper 1 Q19)

- A) $\frac{5\sqrt{7}}{16}$
 B) $\frac{3\sqrt{14}}{16}$
 C) $\frac{\sqrt{7}}{4}$
 D) $\frac{3}{4}$



Answer: A) $\frac{5\sqrt{7}}{16}$

Method: Let $AC = b$. The cosine rule gives $36 = 64 + b^2 - 16b \cos \theta$, i.e. $b^2 - 16b \cos \theta + 28 = 0$, whose two roots $b_1 > b_2$ are the two possible lengths of AC . Both triangles have area $\frac{1}{2} \cdot 8 \cdot b \sin \theta$, so $b_1 = 4b_2$. The product of the roots is $28 = 4b_2^2$, so $b_2 = \sqrt{7}$ and $b_1 = 4\sqrt{7}$; their sum is $5\sqrt{7} = 16 \cos \theta$, so $\cos \theta = \frac{5\sqrt{7}}{16}$. Option B takes the area ratio as the square of the length ratio ($b_1 = 2b_2$); option C uses b_1 alone as the sum of the roots.

Investigate further: For which angles θ do the data $AB = 8$, $BC = 6$ fit two triangles at all? Check that $\sin \theta = \frac{9}{16}$ here is in that range.

6. The circles $C_1 : (x + 3)^2 + (y - 1)^2 = 25$ and $C_2 : (x - 8)^2 + (y - 1)^2 = 9$ have radii 5 and 3. How many common tangents do they have?
- A) 0
 B) 1
 C) 2

D) 3

E) 4

Answer: E) 4

Method: Centres $(-3, 1)$ and $(8, 1)$, so $d = 11$. Since $d > r_1 + r_2 = 8$, the circles are separate and have 4 common tangents: 2 direct and 2 transverse.

Investigate further: Keep C_1 fixed and slide the centre of C_2 along $y = 1$. For which centres are there exactly 3 common tangents? Exactly 1?

7. The point P moves so that $PA = 3PB$, where $A = (0, 0)$ and $B = (6, 0)$. What is the greatest possible value of PA ?

A) 9

B) 7.5

C) 12

D) 6

Answer: A) 9

Method: The locus is a circle symmetric about the x -axis, and A lies on that axis, so the point of the circle farthest from A is where the axis crosses the circle on the far side. The crossings solve $|x| = 3|x - 6|$: $x = \frac{9}{2}$ and $x = 9$. So the greatest PA is 9, at $(9, 0)$ where $PB = 3$. (The circle is $(x - \frac{27}{4})^2 + y^2 = (\frac{9}{4})^2$.)

Investigate further: What is the least possible value of PB ?

8. Fil finds the locus of P with $PA = 3PB$, where $A = (0, 0)$ and $B = (2, 0)$, by writing $(x - 2)^2 + y^2 = 9$. What is wrong?

A) Squaring loses sign information, so the locus is really a single point.

B) The locus of $PA = 3PB$ is a straight line, not a circle.C) His equation says $PB = 3$; it never uses the distance to A .D) Nothing: $(x - 2)^2 + y^2 = 9$ is the locus.**Answer:** C)

Method: $(x - 2)^2 + y^2 = 9$ is the circle with centre B and radius 3: it says $PB = 3$. The true locus comes from $x^2 + y^2 = 9((x - 2)^2 + y^2)$, i.e. $8x^2 - 36x + 8y^2 + 36 = 0$, which is $(x - \frac{9}{4})^2 + y^2 = \frac{9}{16}$: centre $(\frac{9}{4}, 0)$, radius $\frac{3}{4}$. Option A is wrong because both sides are lengths, so squaring loses nothing; option B is true only when the ratio is 1.

Investigate further: Show that A and B are inverse points in the true circle: with centre $C = (\frac{9}{4}, 0)$, check that $CA \cdot CB = r^2$.

Section D Challenge Ramp

1. The locus of P with $PA^2 + PB^2 = 34$, where $A = (-3, 0)$ and $B = (3, 0)$, is a circle. What is its radius?

A) $2\sqrt{2}$

B) 4

C) $\sqrt{17}$

D) $\sqrt{34}$

Answer: A) $2\sqrt{2}$

Method: $PA^2 + PB^2 = (x+3)^2 + y^2 + (x-3)^2 + y^2 = 2x^2 + 2y^2 + 18 = 34$, so $x^2 + y^2 = 8$: a circle about the midpoint of AB with $r = 2\sqrt{2}$. Check: $PA^2 + PB^2 = 2OP^2 + \frac{1}{2}AB^2 = 16 + 18 = 34$. Option C forgets the 18; option D takes 34 as r^2 .

Investigate further: For which values of the constant c does $PA^2 + PB^2 = c$ describe a circle, a single point, or no points at all?

2. S is the square with vertices $(0,0)$, $(4,0)$, $(4,4)$ and $(0,4)$. What is the area of the region of points whose distance from the boundary of S is at most 1?

A) $32 + \pi$

B) $28 + 4\pi$

C) $23 + \pi$

D) $28 + \pi$

Answer: D) $28 + \pi$

Method: Outside S : four 4×1 strips along the sides (16) and four quarter-discs of radius 1 at the corners (π). Inside S : every point except those more than 1 from all four sides, which form the 2×2 square in the middle, so $16 - 4 = 12$. Total $28 + \pi$. Option A forgets the hole; option B uses whole discs at the corners; option C takes the inner square as 3×3 .

Investigate further: Replace the square by a circle of radius 4. Show that the area within 1 of the circle is exactly $2 \times$ its perimeter, but for the square it is less. Where does the square lose area?

3. The diagram shows a kite $PQRS$ whose diagonals meet at O , with $OP = OR = x$, $OQ = y$ and $OS = z$. Which condition is necessary and sufficient for P , Q , R and S to lie on a circle?
(after TMUA 2022 Paper 2 Q11)

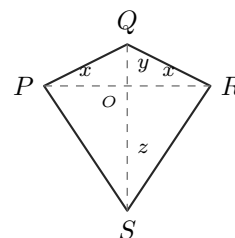
A) $x = y = z$

B) $2x = y + z$

C) $x^2 = yz$

D) $y = z$

E) $y^2 = x^2 + z^2$



Answer: C) $x^2 = yz$

Method: The kite is symmetric in QS , so $\angle SPQ = \angle SRQ$. It is cyclic iff these opposite angles sum to 180° , i.e. iff $\angle SPQ = 90^\circ$. With O at the origin, $P = (-x, 0)$, $Q = (0, y)$, $S = (0, -z)$, and $\vec{PQ} \cdot \vec{PS} = x^2 - yz$, so the condition is $x^2 = yz$. Option A (a square) is sufficient but not necessary; option D makes a rhombus, which is cyclic only when it is a square.

Investigate further: Get $x^2 = yz$ in one line from the intersecting chords theorem applied to PR and QS . Why does that argument also prove the converse?

4. The circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x-4)^2 + y^2 = 9$ meet at P and Q . What is the length PQ ?

- A) 3
 B) 6
 C) $\frac{4\sqrt{14}}{3}$
 D) $\frac{24}{5}$

Answer: B) 6

Method: Subtracting the equations gives the radical axis: $x^2 - (x - 4)^2 = 16$, so $8x - 16 = 16$ and $x = 4$. It is 4 from the centre of C_1 , so the half-chord is $\sqrt{25 - 16} = 3$ and $PQ = 6$. The line $x = 4$ passes through the centre of C_2 , so PQ is also a diameter of C_2 . Option A is the half-chord.

Investigate further: Find the cosine of the angle between the radii O_1P and O_2P .

5. The points A and B are fixed, and $k > 0$ with $k \neq 1$. Prove that the locus of points P with $PA = kPB$ is a circle, and that this circle cuts every circle through A and B at right angles.

Answer: Proof: see method

Method: Similarities preserve circles, ratios of lengths and right angles, so take $A = (-1, 0)$ and $B = (1, 0)$. Both PA and PB are non-negative, so $PA = kPB$ is equivalent to $PA^2 = k^2PB^2$: $(x + 1)^2 + y^2 = k^2((x - 1)^2 + y^2)$, i.e. $(1 - k^2)(x^2 + y^2) + 2(1 + k^2)x + (1 - k^2) = 0$. Since $k \neq 1$ we can divide by $1 - k^2$: $x^2 + y^2 + 2\lambda x + 1 = 0$ with $\lambda = \frac{1+k^2}{1-k^2}$. For $k > 0$, $|1 + k^2| > |1 - k^2|$, so $|\lambda| > 1$ and $r^2 = \lambda^2 - 1 > 0$: the locus is the whole circle with centre $(-\lambda, 0)$. A circle through A and B has its centre on their perpendicular bisector $x = 0$, so its equation is $x^2 + y^2 + 2fy + e = 0$, and passing through $(1, 0)$ gives $e = -1$. Circles $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ are orthogonal iff $2g_1g_2 + 2f_1f_2 = c_1 + c_2$ (Worksheet #3). Here $g_1 = \lambda$, $f_1 = 0$, $c_1 = 1$ and $g_2 = 0$, $f_2 = f$, $c_2 = -1$: the left side is 0 and the right side is $1 - 1 = 0$. So the Apollonius circle is orthogonal to every circle through A and B . ■

Investigate further: Where does the circle cross the line AB ? Show that those points divide AB internally and externally in the ratio $k : 1$.

Top 5 Patterns — Worksheet #7

- $PA = kPB$ with $k \neq 1$ is a circle.** Its crossings with line AB divide AB internally and externally in the ratio $k : 1$; they are the ends of a diameter, so the centre is their midpoint. (see A3, B1, B6, C1, C7, C8, D5)
- Name the locus from the condition.** Fixed distance from a point: circle; from a line: two lines; equal distances: a bisector; fixed angle on a fixed chord: an arc; $PA^2 + PB^2$ fixed: a circle about the midpoint; distance from a shape: a band. (see A1, A2, A4, A5, A10, B2, B4, C2, D1, D2)
- Necessary and sufficient are two separate checks.** Every example must have a necessary property; a sufficient one forces the conclusion. One counterexample kills one direction. (see A9, B5, B7, B8, C4, D3)
- A proof must cover every configuration.** Check where the centre, a point or a second triangle can go; SSA data can fit two triangles, and a figure can show only one case. (see C3, C5, C8)
- Circle equations: complete the square, subtract for chords, compare d with the**

radii.

(see A6, A7, A8, B3, B9, B10, C6, D4)

Common Traps — Worksheet #7

1. **Centring an Apollonius circle at B or at the midpoint of AB .** Neither is right unless you have done the algebra; use the two axis crossings. (see B1, C1, C7, C8)
2. **Keeping one branch of a locus.** $|y - 4| = 3$ is two lines, $|x + y| = |x - y|$ is two lines, and a fixed angle on AB gives an arc on each side. (see A4, B4, C2)
3. **Swapping necessary and sufficient.** Equal diagonals are necessary for a rectangle but not sufficient; a square kite is cyclic, but cyclic kites need not be squares. (see B5, B7, B8, D3)
4. **Accepting a proof because its conclusion is true.** A true claim can have a proof that misses a case or proves something else. (see C3, C8)
5. **Mixing up lengths, squares and areas.** Triangles sharing an angle and a side have areas in the ratio of their other sides, not its square; $PA^2 + PB^2 = 34$ does not make $r^2 = 34$; and a band around a shape has corners and a hole to account for. (see C5, D1, D2)

“A locus is a sentence the entire family agrees on; a geometric claim is only as good as the condition you actually invoked.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths